

19. Risk Prediction in Chronic Kidney DiseaseaaThe authors would like to thank Mackenzie Alexiuk for her invaluable editorial assistance throughout this chapter_ . asse

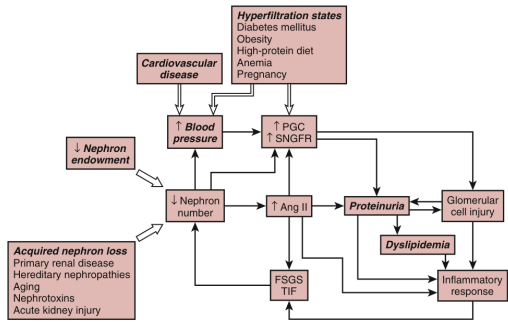


Fig. 19.1 Schematic representation showing the interaction of risk factors for chronic kidney disease progression with pathophysiologic mechanisms that contribute to a vicious cycle of progressive nephron loss. *Ang II*, Angiotensin II; *FGSG*, focal segmental glomerulosclerosis; *PGC*, glomerular capillary hydraulic pressure; *SNGFR*, single-nephron glomerular filtration rate; *TIF*, tubulointerstitial fibrosis. (Adapted from Taal MW, Brenner BM. Predicting initiation and progression of chronic kidney disease: developing renal risk scores. *Kidney Int.* 2006;70:1694–1705.)

Fig_19_1

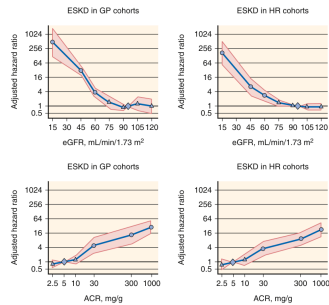


Fig. 19.4 Pooled hazard ratios [95% confidence interval] for end-stage kidney disease (ESKD) according to spline estimated glomerular filtration rate (eGFR) (upper panels) and albumin-to-creatinine ratio (lower panels), adjusted for each other and for age, gender, and cardiovascular risk factors (continuous analyses). Reference categories are eGFR of 95 mL/min/1.73 m² and albumin-to-creatinine ratio of 1 mg/g or diastolic negative of risk. (Left panels) Results for general population cohorts. (Right panels) High-risk cohorts. Dots represent statistical significance, triangles represent nonsignificance, and shaded areas are 95% confidence interval. ACR, Albumin-to-creatinine ratio; GP, general population; HR, hazard. From Gansevoort RT, Matsushima K, van der Velde M, et al. Chronic Kidney Disease Prognosis Consortium: lower estimated GFR and higher albuminuria are associated with adverse kidney outcomes in both general and high-risk populations. A collaborative meta-analysis of general and high-risk population cohorts. *Kidney Int*. 2011;79:1341–1352.)

Fig_19_4

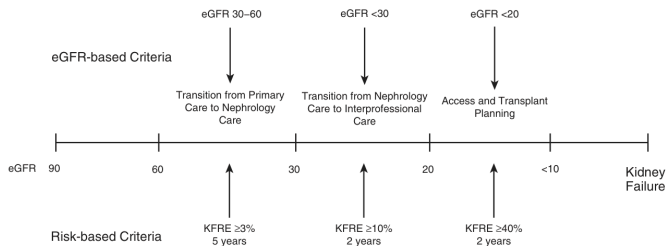


Fig. 19.7 A risk-based versus estimated glomerular filtration rate (eGFR)-based approach to clinical decision making in patients with chronic kidney disease (CKD). *KFRE*, Kidney failure risk equation. (Reproduced with permission from Tangri N, Ferguson T, Komenda P. Pro: risk scores for chronic kidney disease progression are robust, powerful and ready for implementation. *Nephrol Dial Transplant*. 2017;32:748–751.)

Fig_19_7

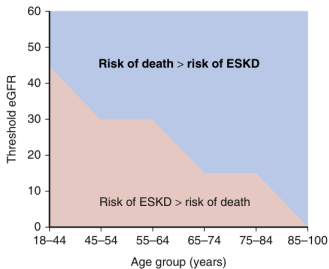


Fig. 19.2 Basene estimated glomerular filtration rate (eGFR) threshold. Below this, the risk for end-stage kidney disease (ESKD) exceeded the risk for death in each age group among 209,622 U.S. veterans with chronic kidney disease stages 3 to 5 followed for a mean of 3.2 years. (From O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol* 2007;18:2758–2765.)

Fig_19_2

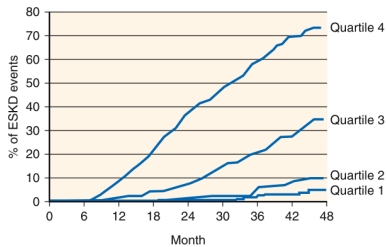


Fig. 19.5 Kaplan-Meier curve for the end-stage kidney disease (ESKD) endpoint stratified by quartile of risk score in 1513 patients with diabetic nephropathy from the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) study. *NIDDM*, Non-insulin-dependent diabetes mellitus. (From Keane WF, Zhang Z, Lyle PA, et al. Risk scores for predicting outcomes in patients with type 2 diabetes and nephropathy: the RENAAL study. *Clin J Am Soc Nephrol*. 2006;1:761-767.)

Fig_19_5

Parameter	Explanation
Strength of association	The stronger the association, the more likely the relationship is causal.
Consistency	A causal association is consistent when replicated in different populations and studies.
Specificity	A single putative cause produces a single effect.
Temporality	Exposure precedes outcome (i.e., risk factor predates disease).
Biological gradient	Increasing exposure to risk factors increases risk of disease, and reduction in exposure reduces risk.
Plausibility	The observed association is consistent with biologic mechanisms of disease processes.
Coherence	The observed association is compatible with existing theory and knowledge in a given field.
Experimental evidence	The factor under investigation is amenable to modification by an appropriate experimental approach.
Analogy	An established cause-and-effect relationship exists for a similar exposure or disease.

Modified from Hill AB. The environment and disease: association or causation? *Proc R Soc Med*. 1956;58:295-300.

Table_19_1

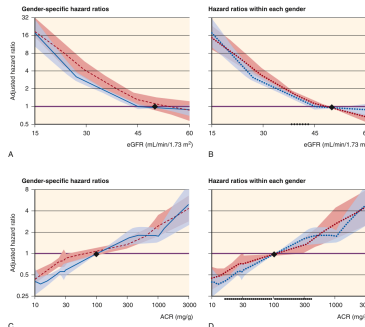


Fig. 19.3 (A and B) Hazard ratios of end-stage kidney disease according to estimated glomerular filtration rate (eGFR). (C and D) Urinary albumin-to-creatinine ratio (ACR) in men versus women in chronic kidney disease cohorts. (A and C) Gender-specific hazard ratios including a main effect for male gender at the reference point. (B and D) Hazard ratios in each gender, thus visually removing the baseline difference between men and women. Hazard ratios were adjusted for age, gender, race, smoking status, systolic blood pressure, history of cardiovascular disease, diabetes, serum total cholesterol concentration, body mass index, and estimated glomerular filtration rate splines or albuminuria. CI, Confidence interval. (Adapted from Nitsch D, Grams M, Sang Y, et al. Associations of estimated glomerular filtration rate and albuminuria with mortality and renal failure by age: a meta-analysis. *BMC*. 2013;346:1324.)

Fig_19_3

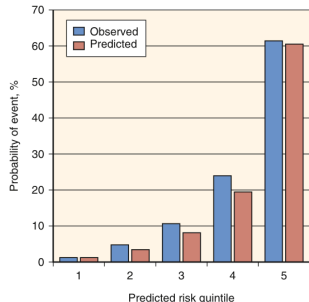


Fig. 19.6 Renal risk score predicted versus observed risk of developing end-stage kidney disease at 3 years in a validation cohort of patients with chronic kidney disease stages 3 to 5. (From Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA*. 2011;305:1553–1559.)

Fig_19_6

Table 19.2 Risk Factors for Chronic Kidney Disease			
Risk Factor	Susceptibility	Initiating	Progression
Older age	+		
Sex	+		
Ethnicity			+
Family history of chronic kidney disease	+		
Metabolic syndrome	+		
Hemodynamic Factors			
Low nephron number			+
Diabetes mellitus	+		+
Hypertension		+	
Obesity	+		
High protein intake	+		
Pregnancy		+	
Primary renal disease			+
Genetic renal disease			+
Urologic disorders		+	
Acute kidney injury		+	
Cardiovascular disease			+
Albuminuria			+
Hypoalbuminemia			+
Anemia	+		
Dehydration	+		
Hypervolemia	+		
Asymptomatic dimethylarginine			+
Hypersphosphatemia			+
Low serum bicarbonate			+
Smoking	+		
Neutrophils	+		

Table_19_2

